

Temporal Bell Inequalities and Global Consistency in Modal Triplet Theory

A Conditional Sequential-Instrument Analysis

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Abstract

We give a typed analysis of Leggett–Garg inequalities in Modal Triplet Theory (MTT). Three distinctions are decisive. First, the laboratory measurement times $t_1 < t_2 < t_3$ belong to a supplied physical clock on a causal spacetime; they are not the auxiliary stabilization parameter used to construct a fixed point. Second, pairwise temporal correlations are generated by declared channels and sequential measurement instruments, including their state-update maps. Third, a Leggett–Garg violation excludes a single context-independent joint distribution for the measured pair protocols; it does not identify projection alone as the cause.

We prove the standard bound $-3 \leq K \leq 1$ from the common-joint-distribution contract. We prove conditional no-retro-signaling for a causal sequential instrument model and distinguish it from noninvasive measurability and no-signaling in time. We also prove a non-selection result: projection or global-history language without dynamics, instruments, and a probability source supports both Leggett–Garg-satisfying and violating realizations. Finally, an explicit qubit, unitary, and Lüders-instrument packet gives $K = 3/2$. This is a standard quantum operational realization. It becomes an MTT result only if a selected MTT source descends to that same packet. Thus global consistency remains a possible mechanism for temporal contextuality, but the present paper is a conditional reconstruction and completion contract, not a generic MTT prediction of temporal Bell violation.

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1 Scope and Claim Tier

Leggett–Garg inequalities test a conjunction of assumptions about a single system measured at different physical times [1]. In their usual macrorealist interpretation, macrorealism per se is combined with noninvasive measurability and an induction or causal-order assumption. Experimental interpretation must also control the measurement-invasiveness or “clumsiness” loophole [3, 2].

The exact contribution of this paper is narrower:

1. identify the common-joint-distribution assumption that implies the three-time bound;
2. define the channels and instruments that generate the measured correlations;
3. state sufficient conditions for operational no-retro-signaling;
4. show what a selected MTT realization would have to emit;
5. exhibit the standard qubit benchmark.

The paper does not derive a Born state, a measurement instrument, physical time, a causal channel, or an objective history-selection law from projection alone. The current MTT shadow-bridge result likewise treats instruments as selected updates rather than inverse or projection maps [6].

2 Two Parameters That Must Not Be Identified

Definition 1 (Physical measurement time). Let Y_4 be a supplied time-oriented Lorentzian space-time, or let a declared laboratory clock provide an equivalent ordered parameter. The events

$$t_1 < t_2 < t_3$$

are physical measurement times on that clock. They label preparation, propagation, coupling, and readout events.

Definition 2 (Stabilization parameter). An auxiliary parameter $\tau \geq 0$ may label an iteration or semiflow

$$\frac{du}{d\tau} = -\mathcal{G}(u)$$

used to construct a fixed point, equilibrium, or coherent representative. It is a mathematical evolution parameter on the construction space.

Remark 1 (Type separation). No equality $t = \tau$ is assumed. A theorem may use $\tau \rightarrow \infty$ to construct data that are then assigned to a physical experiment at finite clock times t_i . Such a construction does not make stabilization time a physical clock, a Lorentzian coordinate, or the shared circle phase. This is the same separation enforced in the corrected MTT kinematics program [7].

3 Sequential Operational Data

3.1 Channels and instruments

Let $\mathcal{H}$ be a complex Hilbert space and let $\mathfrak{T}(\mathcal{H})$ be its trace-class operators. A state is a positive $\rho \in \mathfrak{T}(\mathcal{H})$ with $\text{Tr } \rho = 1$.

Definition 3 (Instrument). For measurement time t_i and setting x_i , a quantum instrument is a family

$$\left\{ \mathcal{I}_{q_i|x_i}^{(i)} : \mathfrak{T}(\mathcal{H}) \longrightarrow \mathfrak{T}(\mathcal{H}) \right\}_{q_i \in \{\pm 1\}}$$

of completely positive trace-nonincreasing maps such that the nonselective map

$$\mathcal{I}_{x_i}^{(i)} := \sum_{q_i = \pm 1} \mathcal{I}_{q_i|x_i}^{(i)}$$

is trace preserving. Outcome probability and conditional state are

$$p(q_i|x_i) = \text{Tr } \mathcal{I}_{q_i|x_i}^{(i)}(\rho), \quad \rho_{q_i|x_i} = \frac{\mathcal{I}_{q_i|x_i}^{(i)}(\rho)}{p(q_i|x_i)}$$

when the denominator is nonzero.

This definition includes both an effect and an update law. A POVM alone does not determine the post-measurement state. Instruments are standard operational objects [4].

Let $\mathcal{E}_{ji}$ be a completely positive trace-preserving channel from time t_i to t_j . In a two-time protocol with common preparation ρ_i , the pair probability is

$$p_{ij}(q_i, q_j|x_i, x_j) = \text{Tr} \left[\mathcal{I}_{q_j|x_j}^{(j)} \left(\mathcal{E}_{ji} \left(\mathcal{I}_{q_i|x_i}^{(i)}(\rho_i) \right) \right) \right]. \quad (1)$$

The measured correlator is

$$C_{ij}(x_i, x_j) = \sum_{q_i, q_j = \pm 1} q_i q_j p_{ij}(q_i, q_j|x_i, x_j). \quad (2)$$

3.2 Pair protocols are contexts

In the standard three-term test, C_{12} , C_{23} , and C_{13} may be estimated in distinct experimental runs. In the C_{13} run, the measurement at t_2 is normally omitted. Therefore the three pair distributions are not automatically marginals of one observed three-time distribution. That common marginal structure is precisely part of the classical contract tested by the inequality.

4 The Leggett–Garg Bound

Definition 4 (Common-joint-distribution contract). Three measured pair distributions satisfy the common-joint-distribution contract when there is one probability distribution

$$\mu(q_1, q_2, q_3), \quad q_i \in \{\pm 1\},$$

whose $(1, 2)$, $(2, 3)$, and $(1, 3)$ marginals equal the three corresponding pair-protocol distributions.

Theorem 1 (Three-time Leggett–Garg bound). *If the measured pair distributions satisfy the common-joint-distribution contract, then*

$$-3 \leq K := C_{12} + C_{23} - C_{13} \leq 1.$$

Proof. For every $(q_1, q_2, q_3) \in \{\pm 1\}^3$, define

$$k(q_1, q_2, q_3) = q_1 q_2 + q_2 q_3 - q_1 q_3.$$

If $q_1 = q_3$, then $k = 2q_1 q_2 - 1 \in \{1, -3\}$. If $q_1 = -q_3$, then $k = 1$. Hence $-3 \leq k \leq 1$ pointwise. Expectation with respect to μ gives the claimed bound, because the pairwise expectations of μ are the measured C_{ij} . $\square$

Corollary 1. *An observed value $K > 1$ rules out the common-joint-distribution contract for the three measured pair protocols.*

Remark 2 (What the violation does not decide). The corollary does not by itself determine whether the failed physical assumption is macrorealism, noninvasive measurability, context independence, preparation equivalence, or another auxiliary premise. That interpretation requires the instruments and protocol controls. Measurement disturbance is therefore not a side issue; it is part of the theorem’s empirical interface.

5 Causal Marginals and Measurement Disturbance

5.1 No-retro-signaling

Definition 5 (Operational no-retro-signaling). For $i < j$, operational no-retro-signaling holds when the earlier marginal

$$\sum_{q_j} p_{ij}(q_i, q_j | x_i, x_j)$$

is independent of the later setting x_j .

Proposition 1 (Conditional no-retro-signaling). *Assume Equation (1), a trace-preserving propagation channel $\mathcal{E}_{ji}$, a complete future instrument $\sum_{q_j} \mathcal{I}_{q_j|x_j}^{(j)}$ for every x_j , and no dependence of the preparation, earlier instrument, or pre- t_j channel on the future setting. Then operational no-retro-signaling holds.*

Proof. Summing Equation (1) over q_j and using trace preservation gives

$$\begin{aligned} \sum_{q_j} p_{ij}(q_i, q_j | x_i, x_j) &= \text{Tr} \left[\mathcal{I}_{x_j}^{(j)} \left(\mathcal{E}_{ji} \left(\mathcal{I}_{q_i|x_i}^{(i)}(\rho_i) \right) \right) \right] \\ &= \text{Tr} \mathcal{I}_{q_i|x_i}^{(i)}(\rho_i), \end{aligned}$$

which is independent of x_j . $\square$

Remark 3. Proposition 1 is conditional on a causal operational model. It is not a theorem that every globally constrained history measure has setting-independent earlier marginals. If a future setting is allowed to change the preparation or the measure assigned to earlier records, the proof does not apply.

5.2 No-signaling in time is different

No-signaling in time (NSIT) asks whether inserting an earlier nonselective measurement leaves a later marginal unchanged:

$$\sum_{q_i} p_{ij}(q_i, q_j | x_i, x_j) = p_j(q_j | x_j).$$

Unlike Proposition 1, NSIT generally fails for invasive measurements because the earlier nonselective instrument can change the state sent forward. Operational no-retro-signaling, NSIT, and macrorealist noninvasive measurability are therefore distinct predicates. Temporal correlations also differ structurally from spacelike Bell correlations because forward signaling and disturbance are physically available [5].

6 What Global Consistency Can and Cannot Supply

6.1 A typed history model

An MTT history realization may be described by:

- a selected upper history space Ω_x for each laboratory setting schedule x ;
- a selected probability state or measure ν_x ;
- record maps $R_{i,x} : \Omega_x \rightarrow \{\pm 1\}$;
- a descent map from upper records to the operational packet $(\rho, \mathcal{E}, \mathcal{I})$.

If different pair protocols are assigned setting-dependent history spaces or measures, their pair distributions need not be marginals of one common μ . This is a possible model of temporal contextuality. It does not, however, force a violation: many context-dependent families still satisfy $K \leq 1$.

Proposition 2 (Projection and global-history non-selection). *A projection map, a fixed-point construction, or the bare assertion that histories are globally constrained does not determine the sign or magnitude of a Leggett–Garg violation.*

Proof. Keep the descriptive projection equal to the identity. One compatible realization assigns one context-independent deterministic value $q_1 = q_2 = q_3 = 1$, giving $K = 1$. A second compatible realization uses the qubit channels and instruments of Section 7, giving $K = 3/2$. The same projection data therefore admit both outcomes. The difference lies in the state, dynamics, instruments, and probability law, none of which is selected by the projection itself. The argument applies a fortiori when only the phrase “global constraint” is supplied. $\square$

Theorem 2 (Conditional MTT temporal-correlation bridge). *Suppose one selected MTT source supplies:*

- (i) *a physical clock and causal order for $t_1 < t_2 < t_3$;*
- (ii) *a normalized state or history measure;*
- (iii) *channels and complete instruments whose descent gives Equation (1);*
- (iv) *common preparation and protocol conventions for all three pairs;*

(v) the causal independence assumptions of Proposition 1;

(vi) pair correlators with $K > 1$.

Then that selected realization violates the common-joint-distribution contract while satisfying operational no-retro-signaling.

Proof. Theorem 1 shows that $K > 1$ excludes the common joint distribution. Proposition 1 proves operational no-retro-signaling from assumptions (iii)–(v). $\square$

Remark 4 (Current status). The theorem is a completion contract. The general MTT projection vocabulary does not yet discharge its state, instrument, and source rows. A selected finite recorder may close a narrower apparatus contract without proving the statement for arbitrary temporal experiments.

7 Explicit Qubit Benchmark

Let

$$\rho_* = \frac{\mathbf{1}}{2}, \quad Q = \sigma_z, \quad P_q = \frac{\mathbf{1} + q\sigma_z}{2}.$$

Use the Lüders instrument

$$\mathcal{I}_q(\rho) = P_q \rho P_q$$

and unitary propagation

$$\mathcal{E}_\delta(\rho) = U_\delta \rho U_\delta^\dagger, \quad U_\delta = \exp\left(-\frac{i\omega\delta}{2}\sigma_x\right).$$

The first outcome is uniform. Conditional on outcome q , the probability of the later outcome r is

$$p(r|q) = \text{Tr}\left(P_r U_\delta P_q U_\delta^\dagger\right) = \frac{1 + qr \cos(\omega\delta)}{2}.$$

Therefore

$$\begin{aligned} C(\delta) &= \sum_{q,r=\pm 1} qr p(q)p(r|q) \\ &= \cos(\omega\delta). \end{aligned} \tag{3}$$

Estimate C_{12} , C_{23} , and C_{13} in separate pair protocols with equal adjacent separation Δ . Equation (3) gives

$$C_{12} = C_{23} = \cos \theta, \quad C_{13} = \cos(2\theta), \quad \theta = \omega\Delta.$$

Hence

$$K = 2 \cos \theta - \cos(2\theta) = 1 + 2 \cos \theta (1 - \cos \theta).$$

At $\theta = \pi/3$,

$$K = \frac{3}{2}.$$

The nonselective Lüders map leaves ρ_* invariant, so the one-time marginals in this benchmark are uniform. The selective instrument still updates the conditional state. The packet is thus explicit enough to evaluate both the correlations and the update law.

Remark 5 (Tier of the benchmark). This calculation is a standard quantum realization, not a derivation of quantum instruments from MTT. It is reproduced by an MTT model only when a selected source and descent theorem emit the same state, channel, instrument, clock convention, and correlators without importing the target values.

8 Relation to Spatial Bell Tests

Spatial Bell and temporal Leggett–Garg inequalities both diagnose failures of declared factorization contracts. The analogy ends there unless a typed intertwiner is supplied. Spacelike Bell protocols impose two-way no-signaling between separated laboratories. Sequential temporal protocols allow forward disturbance and communication through the system. Therefore one cannot transfer Jarrett’s spatial decomposition unchanged to time, nor infer a theorem of spatial–temporal unification from the word “projection.”

Global consistency may provide a common interpretive language for the two domains. A physical unification theorem would additionally require common source data, explicit maps to both operational scenarios, preservation of their probability laws, and a proof that the relevant causal constraints commute with both descents.

9 Completion and Falsification Contract

A physical MTT claim about temporal Bell experiments should report:

- T1.** the physical clock and causal channel;
- T2.** the preparation state or normalized history measure;
- T3.** every measurement instrument, including nonselective and conditional updates;
- T4.** which pair protocols are run and which measurements are omitted;
- T5.** the measured or predicted C_{ij} and uncertainty budget;
- T6.** no-retro-signaling and NSIT tests as separate rows;
- T7.** an invasiveness-control protocol;
- T8.** the MTT source path and descent certificate.

The proposal fails at its declared tier if the source does not emit the instrument packet, if protocol-dependent preparations are hidden, if earlier marginals depend on later settings contrary to the claimed causal contract, or if the predicted K and auxiliary disturbance tests disagree with held-out data.

10 Version 2 Revision Record

Version 2 makes the following corrections:

1. physical measurement times are separated from stabilization time;
2. measurement instruments and update rules are explicit;
3. the LGI theorem is stated for the common-joint-distribution contract;
4. violation is not attributed to projection or global consistency alone;
5. no-retro-signaling is conditional on a causal sequential model;
6. NSIT, no-retro-signaling, and noninvasive measurability are separated;

7. the qubit calculation is retained as an imported operational benchmark;
8. the former generic temporal-network and spatial-temporal unification claims are withdrawn.

11 Conclusion

The robust result is the theorem boundary. A common context-independent joint distribution implies $K \leq 1$. Explicit sequential instruments are needed to calculate the pair correlations and to identify disturbance. Causal channels and complete future instruments imply operational no-retro-signaling, but this does not establish noninvasive measurability. The qubit Lüders packet gives $K = 3/2$ exactly.

MTT can host this packet and can use global-history data to model context dependence. It does not predict the violation from projection, fixed points, or global consistency alone. The remaining scientific step is a selected same-source descent from MTT geometry and dynamics to the state, channels, instruments, and uncertainty-controlled observables of a temporal experiment.

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